Self-Build Specification

Shell and Finishing Works — Furniture Excluded

Porch → Mudroom Conversion · Orange (84100), Vaucluse, France · July 2026 · Rev B

SCOPE

New mudroom (interior ≈ 2.0 × 2.5 m) + stair-head landing (≈ 2.0 × 1.4 m): footing, parpaing block walls, beam-and-block floor, partition wall, insulation + linings, renders, paint, tiling and the K1 door installation. Furniture (W1 · B1 · shoe tower · M1) is OUTSIDE this specification — see the Tender File.

WHO IT IS FOR

A semi-skilled owner doing most of the work personally. Every method statement gives: step-by-step sequence + hand tools + most frequent mistakes. Two trades are deliberately left to professionals (Section 4): electrics and the roof-junction waterproofing.

HOW TO USE IT

First approve the material decisions in Sections 1–2, then execute the method statements (MS-01…09) in construction order. Quantities come from the bill of quantities (BoQ); dimensions are nominal and are fixed on measurement day.

STATUS

Rev B — July 2026. Circulation: stair → K1 exterior door → landing → V door → mudroom. The old east (stair-side) doorway is cancelled — walled up.

CONTENTS

SITE MEASUREMENT BEFORE FABRICATION IS MANDATORY — NO DIMENSION GOES TO FABRICATION.

Read together with the BoQ, the A3 drawing pack and the K1 door datasheet — not a fabrication document on its own.

1 Floor decision — material selection

Mudroom + landing floor ≈ 8 m² (tile requirement incl. waste ≈ 8.8 m²). The floor sees wet, muddy shoes every day, and the K1 threshold borders the outside climate. Three candidate materials were compared on five criteria; the decision sits below the table.

Criterion	Marble	Porcelain stoneware (grès cérame)	Ceramic (glazed)
Slip safety — wet shoes (target R10–R11)	Polished surface is slippery when wet; usually no R-class certificate	R10–R11 certified ranges are plentiful — the right class for an entrance	Most ranges are R9; anti-slip certificate not universal
Freeze–thaw (around the K1 threshold)	Porous; sensitive to frost and the salt–mud cycle	Water absorption ≤ 0.5% — frost-proof, used even on external thresholds	High water absorption (> 3%) — cracking risk at the threshold
Maintenance	Impregnation/polish once a year; acid cleaners forbidden, stains easily	Wipe with neutral detergent — needs nothing else	Scratched glaze stains; joints soil quickly
Material cost (€/m², order of magnitude)	≈ €40–90 (+ cutting/polishing labour)	≈ €20–45	≈ €10–25
DIY laying difficulty (1–5)	4–5 — heavy, hard to cut, mortar-stain risk	2–3 — 60×60 lays well with double buttering (MS-07)	2 — the easiest, but the wrong class

CLEAR RECOMMENDATION — R10 PORCELAIN STONEWARE 60×60 (GRÈS CÉRAME), MATT/TEXTURED

In an entrance used daily with wet shoes, slip resistance is only guaranteed by a certificate; R10 porcelain carries that certificate as standard, whereas marble usually has no such class at all. With water absorption ≤ 0.5% it withstands the freeze–thaw and mud–salt cycle around the K1 threshold with zero maintenance. The 60×60 format reduces joints over the ≈ 8.8 m² area and is the most economical safe option a semi-skilled builder can lay personally with the MS-07 method (≈ €20–45/m²).

- Purchasing: buy all tiles from the SAME calibre/shade batch; ≈ 8.8 m² incl. 10% waste (same as the BoQ).
- On the external threshold strip right in front of K1, the same range's R11 textured outdoor tile can be used.
- Skirting: on furniture-free wall faces, a band h ≈ 70 mm cut from the same tile (MS-07, step 7).
- Laying method, joints and perimeter mastic: MS-07. Screed level is set by deducting the tile thickness (MS-03, step 7).

LEVEL REMINDER

The target finished floor level is the SAME on landing and mudroom: +1.25 m — a threshold-free passage at the V door. At the K1 threshold a single step of at most 20 mm. Remember to deduct tile + adhesive thickness from the screed level.

2 Wall decision — wall by wall

The rule is simple: the inner faces of new masonry walls get insulation + plasterboard lining (doublage); the landing–mudroom partition is metal studs + double plasterboard; ALL interior surfaces finish with washable matt paint. The table below settles every wall one by one.

Wall	Build-up	Inner face lining	Note
MUDROOM · East	New parpaing 20 + glazing band 1000×450 in the top strip (sill +2000 — ASSUMPTION); the old doorway is walled up in Rev B	Doublage: 100 mm wool (R ≥ 3.0) + BA13 → filler + paint (MS-05)	B1 bench wall; glazing-band edges with corner beads
MUDROOM · North	New parpaing 20	Doublage — same system	M1 mirror unit on this wall; recessed mirror on the V door axis (x = 870)
MUDROOM · West	Existing masonry 200	Doublage — the existing wall is insulated too (same system)	W1 wardrobe wall; lining + paint finished BEFORE furniture installation
MUDROOM · South (= partition)	Light partition: metal studs + 2 faces × 2 layers BA13 + 45 mm wool; contains the V door 730×2100 (timber, east hinge)	Both faces filler + paint (MS-04)	Shoe tower on the mudroom face of the partition, east end
LANDING · East	New parpaing + K1 door (900×2100 alu)	Doublage on the inner face	K1 threshold/seal detail: MS-08
LANDING · South + West	Existing walls (hall and cellar doors outside scope, unchanged)	Local repair filler + paint	Architrave feet re-sealed with mastic

PAINT — ONE SPECIFICATION

All interior surfaces: plasterboard primer + 2 coats of WASHABLE MATT interior paint (mat lessivable). No gloss/semi-gloss (shows defects); no cheap non-washable matt (an entrance takes stains). Apply at ≥ 10 °C — MS-06.

DAMP 1 — AROUND THE THRESHOLDS

Around the K1 and V thresholds the bottom band (h ≈ 300 mm) uses water-resistant GREEN (hydrofuge) BA13; under the threshold an EPDM gasket + sealing tape (MS-08). Tile skirtings at the wall feet stop splash water.

DAMP 2 — EXTERNAL CORNER BEADS

Every external corner and every door/glazing-band edge gets a mesh-winged PVC/alu corner bead — inside under the filler (a straight line + impact protection), outside within the monocouche render (MS-09). Without a bead the corner filler cracks within 6 months; in an entrance the corners take luggage and shoe knocks.

DRAWING NOTE — WHY THE V DOOR OPENS INTO THE MUDROOM (HINGE WEST)

The landing is shallow (depth 1400 — ASSUMPTION); if V swung into the landing the leaf would block the passage. So V opens INTO THE MUDROOM; the hinge is on the WEST jamb, the leaf rests against W1's side panel and keeps the entry axis clear — whoever enters sees the mirror straight ahead. Even if K1 and V open at the same time the leaves stay in different volumes; there is no clash scenario. (Source: dimensions.json · swing_rationale)

MS-01 Method statement — Footing + strip foundation (fondation filante)

A strip footing under the new east and north walls: section 500 × 250 mm, base below frost depth (≥ 600 mm below finished ground), concrete C25/30, reinforcement 4×HA10 + stirrups.

MATERIALS (SAME AS THE BoQ): concrete C25/30 ≈ 0.85 m³ (footing) · rebar HA10 ≈ 29 kg (incl. ring beam) · plastic cover spacers · cement + sand for the blinding layer

SEQUENCE OF WORK

1. **Setting out**
   Mark the wall axes with pegs + string; check the corner square with a 3-4-5 triangle. Trench width 500 mm.
2. **Excavation**
   Footing base on natural ground and ≥ 600 mm deep. Keep the base level; remove loose soil and compact it with a hand rammer.
3. **Blinding**
   40–50 mm of lean concrete — keeps the rebar out of the mud and gives a flat working level.
4. **Reinforcement**
   4×HA10 longitudinal + stirrup cage; cover ≥ 40 mm (plastic spacers), laps ≥ 500 mm and wired.
5. **Pour**
   Pour the C25/30 in one go; rod it with a rebar, strike the top to the string/laser level with a trowel.
6. **Curing**
   Keep moist for 7 days (cover + wetting). Start blockwork after 3 days at the earliest; full strength at 28 days.

HAND TOOLS

- pick + shovel + wheelbarrow
- tape measure, string, timber pegs
- spirit level + laser level (hired)
- rebar cutter + tying pliers
- trowel + hand rammer
- concrete mixer (hired) — or ready-mix (truck)

MOST FREQUENT MISTAKES — AND HOW TO AVOID THEM

1. **Pouring onto loose fill**
   The footing settles, the wall cracks. Take the base to natural ground and compact.
2. **Rebar sitting in the mud**
   Without cover the steel rusts. Use plastic spacers; steel must not touch soil.
3. **Pouring in several batches**
   A cold joint = a weak section. The volume is small (< 1 m³): pour in one go with a helper.
4. **No curing in hot weather**
   Surface dusting + shrinkage cracks. Cover and wet for 7 days.
5. **Leaving the level for later**
   A correction screed becomes necessary. Strike to the string/laser level during the pour.

MS-02 Method statement — Blockwork (parpaing)

The east + north walls and the window/vent infills are built in concrete block (parpaing 20). The old east doorway is CANCELLED in Rev B — walled up with the same method and included in the full wall take-off.

MATERIALS (SAME AS THE BoQ): parpaing 20 ≈ 325 pcs (≈ 29.2 m² of wall) · mortar ≈ 28 bags of 25 kg · arase étanche DPC strip · stainless ties + chemical anchors · lintels · ring-beam U-blocks + 4×HA10 (concrete ≈ 0.27 m³)

SEQUENCE OF WORK

1. **Damp-proof course**
   Laid on the footing BEFORE the first mortar (arase étanche) — the only insurance against rising damp.
2. **Dry run**
   Lay the first course without mortar; plan the cuts and the K1/glazing-band openings.
3. **Corner-to-corner line**
   Build the corner blocks plumb and level, stretch a line between them; every block sits to the line.
4. **Laying**
   Bed joints 10–15 mm, vertical joints FILLED; courses half-block staggered. Damp the blocks on hot days.
5. **Tie to the existing wall**
   Every 2nd course a stainless angle tie + chemical anchor.
6. **Openings**
   Lintels over K1 and the glazing band; bearing ≥ 200 mm each side.
7. **Ring beam (chaînage)**
   Top course U-blocks + 4×HA10 + concrete. Do NOT mortar the wall to the roof timbers — leave compressed mineral wool + a flexible mastic joint.
8. **Checks**
   Every 2–3 courses check plumb (level) and line (straightedge).

HAND TOOLS

- trowel + mortar tub + mixer
- line + corner pins
- spirit level 60–120 cm + straightedge 2 m
- rubber mallet
- Ø230 masonry disc (angle grinder — hired)
- tape measure + chalk

MOST FREQUENT MISTAKES — AND HOW TO AVOID THEM

1. **Forgetting the DPC strip**
   Ground damp climbs the wall and moulds inside. Not a single course without it.
2. **Leaving vertical joints empty**
   Leaks air/water and weakens the wall. Every vertical joint filled with mortar.
3. **Laying by eye, without a line**
   A wavy face = thick, expensive render. A line on every course.
4. **Short lintel bearing**
   < 200 mm bearing cracks the corner. ≥ 200 mm both sides.
5. **Filling the wall–roof joint rigid**
   Timber moves, the wall cracks. Wool + mastic; flashing goes to the pro (Section 4).

MS-03 Method statement — Beam-and-block floor + screed

The floor is raised ≈ 0.90 m on sleeper walls with a poutrelle + hourdis (beam-and-block) system; a ventilated void remains below. Insulation + screed go on top. Target finished level: +1.25 m on landing and mudroom.

MATERIALS (SAME AS THE BoQ): poutrelle 8 pcs · hourdis ≈ 8.0 m² · mesh ST25 ≈ 9.2 m² · topping concrete ≈ 0.36 m³ · PU 80–100 mm ≈ 8.0 m² · screed 50–60 mm · 2 vents

SEQUENCE OF WORK

1. **Sleeper walls**
   3 intermediate sleeper walls (axis ≤ 1000 mm); tops laser-levelled in one plane.
2. **Vents**
   Leave 2 opposite ventilation openings — the void below (≥ 400 mm) must breathe.
3. **Joists**
   Place the poutrelles to plan; respect the maker’s bearing dimension and prop mid-span with a temporary prop (étai).
4. **Infill blocks**
   Lay the hourdis between the joists; edge cuts with the masonry disc.
5. **Mesh + conduits**
   ST25 mesh on cover spacers; run the electrical conduits (ICTA) in this layer.
6. **Topping**
   40–50 mm, in one pour, ruled flat. Cure 7 days; remove the props after 7 days at the earliest.
7. **Insulation + screed**
   PU 80–100 mm + vapour barrier + perimeter edge strip; screed 50–60 mm. Screed level = +1.25 m minus tile + adhesive thickness.
8. **Drying**
   The screed dries ≈ 1 week per cm. Foil test before tiling: tape a foil square overnight — a dry underside in the morning means ready.

HAND TOOLS

- laser level (hired)
- trowel + 2 m alu straightedge
- wheelbarrow + mixer
- mesh/rebar cutter
- Ø230 masonry disc
- temporary props (étai — hired)
- fine-tooth saw for the PU boards

MOST FREQUENT MISTAKES — AND HOW TO AVOID THEM

1. **Skipping the vents**
   Condensation in the void → damp floor, smell. 2 vents, opposite, always open.
2. **Pouring without props**
   The joist deflects, the concrete cracks. An étai at mid-span, stays 7 days.
3. **Screed without edge strip**
   Perimeter crack + a sound bridge to the wall. Foam strip along every wall foot.
4. **Forgetting the tile allowance in the screed level**
   A step appears at the V threshold. Finished target +1.25; deduct tile + adhesive.
5. **Tiling onto a wet screed**
   The adhesive fails, tiles lift. The cm-per-week rule + foil test.

MS-04 Method statement — Partition wall (metal studs + double BA13)

MATERIALS (SAME AS THE BoQ): BA13 ≈ 13.9 m² (2 faces × 2 layers) · profiles ≈ 16.5 m · mineral wool 45 mm ≈ 5.0 m² · acoustic tape · V door opening 730×2100

1. **Layout + rails**
   Snap the partition axis on floor/ceiling with a chalk line (total thickness 100 mm); screw the U rails at 60 cm centres with acoustic tape underneath.
2. **Studs**
   C studs at 600 mm centres; DOUBLE studs at the V door opening + a lintel profile above.
3. **Face 1, layer 1**
   Screw the boards with 25 mm screws every ~30 cm; joints must land on studs.
4. **Wool + services**
   Fill with the 45 mm mineral wool (sound); pull any empty conduits through now.
5. **Closing**
   Other face 2 layers + the first face's 2nd layer (35 mm screws). Joints STAGGERED in every layer.
6. **Joints**
   Tape + 2 coats of jointing compound — continues in MS-06.

HAND TOOLS

- utility knife + board rasp
- screw gun (torque-set)
- profile shears
- chalk line + spirit level
- tape measure
- filling-knife set

MOST FREQUENT MISTAKES

1. **Stacking the joints**
   Without stagger between layers a crack line runs the length of the wall.
2. **Overdriving the screws**
   A torn paper face holds nothing. Use a torque-set bit.
3. **A single stud at the door opening**
   The frame vibrates, the corner cracks. Double stud + lintel profile.

MS-05 Method statement — Doublage — insulation + plasterboard lining

MATERIALS (SAME AS THE BoQ): wool 100 mm GR32 (R ≥ 3.0) ≈ 21.3 m² · BA13 lining ≈ 29.3 m² · vapour barrier + tape · profiles

1. **Level**
   Mark the finished-face plane with the laser; total build-up ≈ 120 mm — the room's interior size (2.0 × 2.5) is measured AFTER this lining.
2. **Framing**
   Rail + stud (fourrure/montant) system; floor and ceiling rails plumb to each other.
3. **Wool**
   Batts upright, tight-joined and UNCOMPRESSED; electrical conduits run in the insulation layer.
4. **Vapour barrier**
   On the warm side; laps ≥ 100 mm and taped, socket penetrations taped/grommeted.
5. **BA13**
   Screw the boards; the bottom band around thresholds in GREEN (hydrofuge) board (Section 2, Damp 1).
6. **Joints**
   Tape + compound — MS-06.

HAND TOOLS

- insulation knife
- screw gun
- profile shears
- laser level
- vapour-barrier tape
- filling knife

MOST FREQUENT MISTAKES

1. **A punctured vapour barrier**
   Condensation forms inside the wool. Tape every lap and penetration.
2. **Compressing the wool**
   Squashed wool loses its R-value. Keep the batt thickness.
3. **Remembering conduits too late**
   The lining comes off again. Pull the conduits and photograph before closing.

MS-06 Method statement — Interior filler + paint

Joints + filler on the plasterboard faces, repair filler on the walled-up opening and existing walls; then primer and two coats of washable matt paint. The paint specification is a single type in Section 2 — the whole room is painted with the same product.

MATERIALS (SAME AS THE BoQ): joint tape + jointing compound · fine surface filler · plasterboard primer (sous-couche) · washable matt interior paint (≈ 31 m² walls + 8 m² ceiling, 2 coats)

SEQUENCE OF WORK

1. **Joints**
   Joint tape + 1st coat of compound; screw heads filled too.
2. **2nd coat**
   Spread with a wide knife; when dry, sand with 120–150 mesh.
3. **Repair faces**
   Repair filler on the walled-up doorway and the existing plaster; skim the whole face if needed (ratissage).
4. **Dust + primer**
   Remove the sanding dust completely; prime EVERY surface with plasterboard primer.
5. **Paint**
   Washable matt, 2 coats; full drying between coats + a very light sand. Apply at ≥ 10 °C.

HAND TOOLS

- filling knives 10 / 20 / 30 cm
- sanding block + 120–150 mesh
- dust mask (FFP2)
- roller + tray + extension pole
- cutting-in brush
- masking tape + dust sheets

MOST FREQUENT MISTAKES — AND HOW TO AVOID THEM

1. **Skipping the primer**
   Filled and paper areas absorb paint differently — the wall looks patchy. Primer everywhere.
2. **One thick filler coat**
   It cracks as it dries. 2–3 thin coats, each after the last has dried.
3. **Choosing non-washable paint**
   In an entrance the first mud stain is permanent. Look for “lessivable” on the tin.
4. **Painting over dust**
   Paint will not bond and peels later. Wipe with a damp cloth after sanding.

THE CEILING FINISHES WITH THE SAME SYSTEM

Ceiling: 60–100 mm insulation + BA13 (suspended framing) finished with the same joint–filler–primer–paint chain; ceiling paint also washable matt, white. Mark the sensor-light and LED-driver cable exits BEFORE filling — the electrical connection itself belongs to the electrician per Section 4.

MS-07 Method statement — Tiling + grout

Section 1 decision: R10 porcelain stoneware 60×60. Do not start this statement before the screed is fully dry (MS-03, step 8). Landing + mudroom at the same level, threshold-free passage at the V door.

MATERIALS (SAME AS THE BoQ): R10 porcelain tile 60×60 ≈ 8.8 m² (incl. 10% waste) · flexible adhesive (C2) · grout 2–3 mm · primer (primaire) · spacers + levelling clips · perimeter mastic

SEQUENCE OF WORK

1. **Substrate check**
   Screed dry (foil test), dust-free and primed.
2. **Dry layout**
   Start from the V door axis; cuts go to the wall feet. Lay the first row to a string/laser line.
3. **Adhesive**
   For 60×60, DOUBLE BUTTERING (double encollage): 8–10 mm notched trowel on the floor + a thin coat on the tile back.
4. **Placing**
   Bed with the rubber mallet; spacers + levelling clips; check level while the adhesive is fresh.
5. **Grout**
   After 24 h grout the joints; clean with a damp sponge within 20–30 min.
6. **Perimeter**
   At the wall feet NOT grout but flexible mastic; mastic under the K1 threshold profile.
7. **Skirting**
   On furniture-free faces a skirting h ≈ 70 mm cut from the same tile.

HAND TOOLS

- notched trowel 8–10 mm
- 600 mm tile cutter (carrelette) or an electric wet saw (hired)
- rubber mallet
- 2–3 mm spacers + levelling clips
- sponge + two buckets
- rubber grout float
- spirit level + laser

MOST FREQUENT MISTAKES — AND HOW TO AVOID THEM

1. **Single-side adhesive**
   Voids behind a big tile: corner breaks + hollow tiles. Double buttering is a must.
2. **Laying on a wet screed**
   Tiles lift, the adhesive separates. Start only after a positive foil test.
3. **Filling the perimeter joint rigid**
   Thermal movement cracks it. Wall feet always in flexible mastic.
4. **Starting from a wall edge**
   A sliver cut lands in front of the door. The layout starts on the V door axis.
5. **Delaying the grout clean-off**
   A film sets in the textured R10 surface. Sponge within 20–30 min.

LEVEL CHECK — BEFORE LAYING

Before tiling, compare the landing and mudroom screed levels with the laser at the V opening: the target is the SAME finished level on both sides (+1.25, threshold-free). If the difference is > 5 mm, correct with levelling screed, not with adhesive thickness — a thick adhesive bed collapses and the tile drums.

MS-08 Method statement — K1 door installation (threshold · seals)

K1: glazed, thermal-break aluminium single leaf 900×2100, Uw ≤ 1.6, RC2 lock, self-closing hinges. On the east edge of the landing, at the stair head; hinge on the NORTH jamb, the leaf opens INTO THE LANDING.

MATERIALS (SAME AS THE BoQ): K1 door set (frame + leaf + RC2 lock) · alu thermal threshold profile (≤ 20 mm) · EPDM gasket + under-threshold sealing tape · compriband · low-expansion foam · hybrid (MS-polymer) sealant

SEQUENCE OF WORK

1. **Opening check**
   Frame outer size + 10–15 mm fitting allowance each side; measure both diagonals. Threshold top level: at most 20 mm above finished floor.
2. **Under the threshold**
   Sealing tape + EPDM; outside, a drip-edge threshold with a 2% outward fall.
3. **Frame**
   Fix plumb + level on packers; no screws before the diagonals are equal. Hinge side NORTH.
4. **Fixing + leaf**
   Anchor screws/pattes; hang the leaf, check for rubbing, adjust at the hinges.
5. **Sealing**
   Low-expansion foam around the frame; airtight tape on the INSIDE face, compriband on the OUTSIDE (water-repellent, vapour-open).
6. **Finishing**
   Hybrid sealant around the outside (paintable); interior architrave; adjust the self-closing speed.

HAND TOOLS

- plumb line + spirit level
- drill + anchor set
- plastic packer set
- foam gun (low-expansion)
- sealant gun
- hex-key set (door adjustment)
- tape measure

MOST FREQUENT MISTAKES — AND HOW TO AVOID THEM

1. **Foam bellying the frame**
   The leaf rubs. Brace the frame with opposed packers/spreaders before foaming.
2. **No tape under the threshold**
   The first downpour walks water inside. Tape + EPDM + 2% outward fall.
3. **Not measuring the diagonals**
   The frame goes parallelogram, the lock will not catch. No fixing until both diagonals match.
4. **Sealing the outer joint with silicone**
   Standard silicone cannot be painted and tears off. Use hybrid (MS-polymer) sealant.

DRAWING NOTE — WHY K1 OPENS INTO THE LANDING (HINGE NORTH)

A door opening OUT over the stair pushes the opener into the stair void and the leaf hits anyone coming up — a fall risk. K1 therefore opens INTO THE LANDING; self-closing hinges stop wind-slam. The hinge is on the NORTH jamb: the open leaf rests against the east end of the landing's north wall (no clash — the V door is shifted west), blocking neither the house door nor the passage. Because the façade changes, K1 is added to the DP file. (Source: dimensions.json · swing_rationale)

MS-09 Method statement — External render (enduit monocouche)

The new east + north façades and the walled-up openings are rendered with a one-coat system render (monocouche); scraped finish (gratté). The colour matches the façade tone in the DP/ABF decision — match the sample to the town hall's reply.

MATERIALS (SAME AS THE BoQ): monocouche ≈ 22.2 m² → ≈ 23 bags of 25 kg (ONE batch) · starter profile · mesh-winged external corner + edge beads · render mesh (trame)

SEQUENCE OF WORK

1. **Substrate**
   Joints filled, surface dust-free; pre-dampen the wall in hot weather.
2. **Beads**
   Starter profile at the bottom; mesh-winged beads on every external corner and the K1/glazing-band edges.
3. **Mix**
   Mix with the water ratio on the bag; identical consistency in every batch.
4. **1st pass**
   10–12 mm body coat; embed mesh with ≥ 200 mm laps over existing–new junctions and above lintels.
5. **2nd pass**
   Total 15–20 mm; rule flat with the straightedge.
6. **Scraped finish (gratté)**
   When the surface has dulled and takes no fingerprint, scrape in circles with the gratton.
7. **Curing**
   Protect from sun and wind for 2–3 days; mist lightly if needed.

HAND TOOLS

- steel trowel + plastic straightedge
- scraper (gratton)
- mixer + a clean bucket
- render mesh roll
- profile shears
- sound platform / scaffold (hired)

MOST FREQUENT MISTAKES — AND HOW TO AVOID THEM

1. **Bags from different batches**
   The façade dries in stripes of different tone. All bags in one purchase, same batch.
2. **Applying in a hot, windy midday**
   Fast water loss makes "burn" cracks. Work early morning or evening.
3. **No mesh at the junction**
   A crack along the existing–new wall line. Mesh embedded with ≥ 200 mm laps.
4. **More than 20 mm in one coat**
   The render sags and drops off. Two passes, 15–20 mm total.
5. **Missing the scraping window**
   Too early: it tears; too late: it will not scrape. Test a small area first.

4 Works left to professionals

ELECTRICS — licensed electrician (électricien)

- Domestic electrics in France fall under standard NF C 15-100; new circuits or board alterations may require a Consuel conformity certificate (attestation). Uncertified work causes trouble with insurance, resale and any claim.
- Everything at 230 V — sockets, lighting circuit, board connection, earthing — goes to a licensed electrician.
- What you can do yourself: place the empty ICTA conduits and box positions before the walls close (MS-03/04/05); connect the 24 V LED strip on the OUTPUT side of the driver. The driver's 230 V supply is the electrician's job.
- Call the electrician twice: at conduit stage (before the linings close) and at completion for connection + test.

ROOF-JUNCTION WATERPROOFING — roofer with décennale insurance

- The waterproofing where the new walls meet the existing roof (flashing / cover profile) is left to a roofer carrying 10-year insurance (garantie décennale).
- The reason is the guarantee: this junction is the riskiest water point of the house. Home insurance can refuse damage caused by self-made flashing; with a professional the 10-year décennale applies — keep the invoice.
- Your part: finish the wall at ring-beam level and leave the wall–roof joint compressible (MS-02, step 7). The roofer executes the flashing detail; just do not make the joint rigid.

5 Close-out — measurement day and document hierarchy

THE MEASUREMENT-DAY RULE

«Site measurement before fabrication is mandatory — nothing may be fabricated from these nominal values.» Every dimension in this specification is nominal (dimensions.json, all verified:false; the landing depth 1400 mm and the glazing-band size are ASSUMPTIONS). When the shell is done the room is laser-surveyed; furniture orders, cutting lists and final door/tile dimensions are issued ONLY after that survey. The take-off (tools/quantities.py) is recalculated after the survey.

THIS SPECIFICATION IS NOT READ ALONE

Documents read together: 1) bill of quantities (BoQ) — binding quantities; 2) A3 drawing pack — plan, sections, elevations; 3) shop drawings S-01…S-05 — furniture (outside this specification's scope); 4) K1 door datasheet (Annex). In case of conflict the precedence is: site measurement → drawing → specification text → BoQ quantity. All documents must be Rev B (July 2026) — a single page from an older revision brings back cancelled details such as the east door.

Read; I approve the material decisions: ____________________ ____________________

Owner — name / signature | Date